CALEB UNIVERSITY, LAGOS · B.Sc. COMPUTER SCIENCE · FINAL YEAR PROJECT

Design and Development of a Nigerian Public Economic Data Aggregation and Analytics Platform with Open API Access

A 2020–2026 Case Study · “NPEDATA”

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THE CONTEXT

The data exists. You just can't use it.

The raw material is reliable; its published form is the problem.

- **Credible sources** — CBN, NBS, World Bank.
- **But fragmented** — scattered PDFs and spreadsheets, published to read, not to compute.
- **High friction** — every question starts with manual cleaning.

THE PROBLEM

Public data, practically closed

Four concrete barriers turn technically-public data into practically-unusable data.

- **Not machine-readable** — locked in PDFs.
- **No common schema** — clashing units and frequencies.
- **No open API** — nothing free to query.
- **Duplicated effort** — everyone re-cleans the same data.

AIM & OBJECTIVES

One store. Two front doors.

Aim Aggregate Nigeria's public economic data into one standardised store, served through a dashboard and a free open API.

1. Collect — CBN, NBS, World Bank into one repository
2. Standardise — one relational model
3. Analyse — change, trend, correlation, server-side
4. Open API — free, documented, HATEOAS, no auth
5. Dashboard — clear, honest visualisations
6. Evaluate — correctness, usability, accessibility

SCOPE

What it is — and isn't

Being explicit about the boundary is deliberate — it pre-empts the hard questions.

- **Is** — a working prototype + open API, 122 indicators.
- **Isn't** — not national infrastructure, not a bank, not real-time.
- **No AI / ML** — transparent by design.
- **Classical analytics** — correlation, trend, forecast.

FOUNDATIONS

Built on established ideas

Every design choice traces to a recognised principle, not personal preference.

- **Open data** — machine-readable means actually usable.
- **Tidy data** — one row per observation (Wickham, 2014).
- **REST & HATEOAS** — a self-describing API (Fielding, 2000).
- **Honest charts** — no dual axes (Tufte; Anscombe).

THE GAP

Nobody puts it all together

Strong systems exist on each side; none combine all three properties at once.

- **Global tools** — FRED, World Bank — thin Nigerian detail.
- **Local sources** — CBN, NBS — no API.
- **Paywalled aggregators** — resell the same figures.
- **The gap** — none combine coverage + open access + honesty.

WHY NOW

Not a technical problem

Every claim here is drawn from published sources, not personal opinion.

- **Framework exists** — Ne-GIF since 2018.
- **Institutional** — data is treated as an asset (Eleanya, 2026).
- **Enforcement gap** — 153 of 250 MDAs fell below the FOI benchmark (Ogunyale & Osho, 2023).
- **This project** — removes the technical excuse.

METHOD

Build in slices, verify each loop

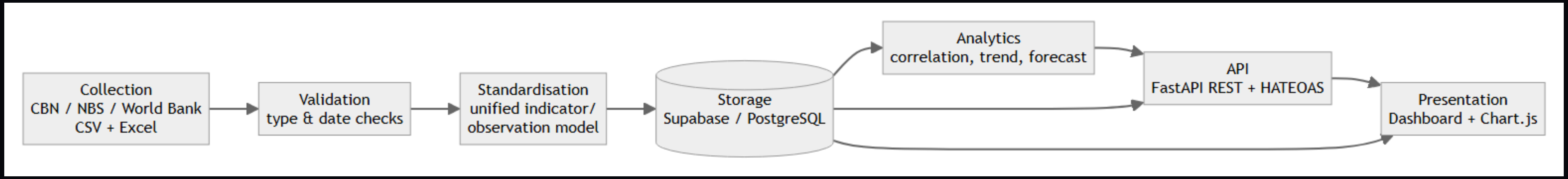
An iterative prototype fits a solo project whose requirements emerged during the build.

- **Iterative prototyping** — model → API → dashboard → analytics.
- **Verify every loop** — figures checked against the database.
- **Not Waterfall or Scrum** — requirements emerged; solo developer.

ARCHITECTURE

One pipeline, seven stages

The core contribution: a pipeline that transforms the data, not a site that re-displays it.



Collect → Validate → Standardise → Store → Analyse → API → Present.

DATA MODEL

One table, every frequency

Separating the value from its metadata lets one table hold every series.

- **Tidy schema** — sources → indicators → observations.
- **One shape** — (indicator, date, value) — daily to annual.
- **Integrity** — keys + uniqueness kill duplicate ingestion.
- **Unit metadata** — no silent ₦'000-vs-₦m errors.

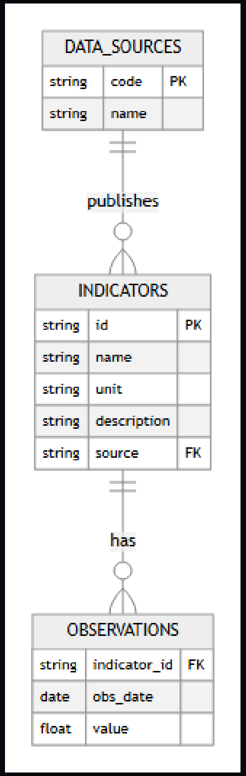


Figure 3.4 — Entity-relationship diagram.

OPEN API

HATEOAS, Level 3

A free, self-describing REST API — the machine access the sources don't provide.

- **FastAPI REST** — 17 endpoints, no key.
- **Self-describing** — every response carries _links.
- **Proven live** — the HATEOAS Explorer follows them.
- **Machine-ready** — MCP + llms.txt for AI agents.

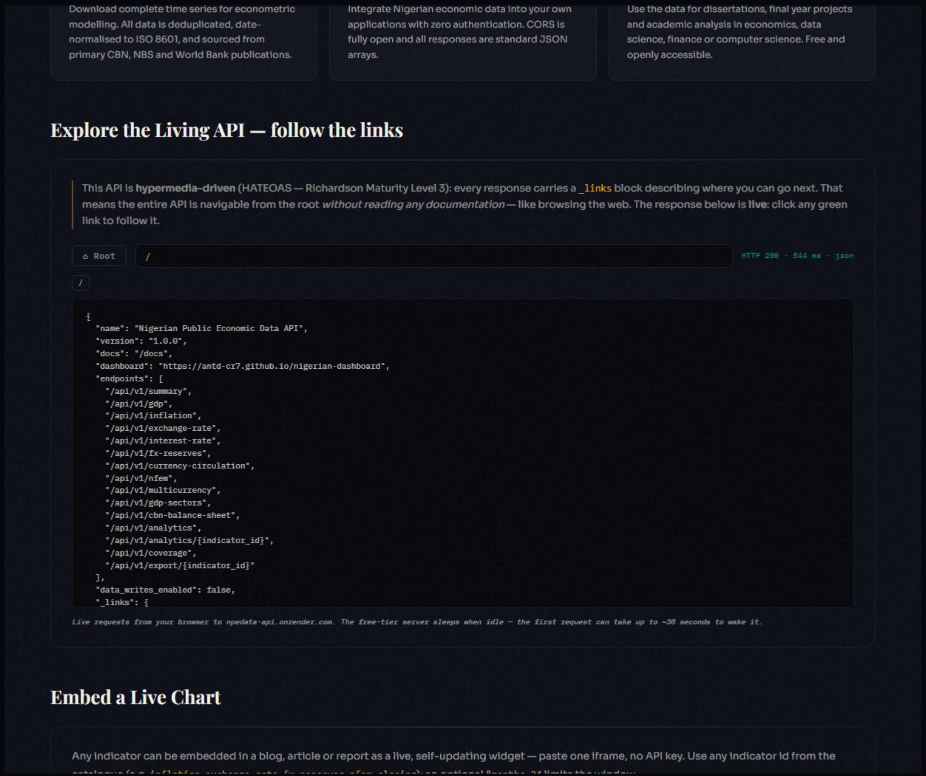


Figure 4.11 — HATEOAS Explorer browsing the live API.

DASHBOARD

One page, two audiences

One honest interface that serves both the public and researchers.

- **Story pattern** — what happened / why / how to read it.
- **Reader ⇄ Analyst dial** — public and researcher, one page.
- **Honest encoding** — no dual axes; units stated.
- **100 / 100** — Lighthouse accessibility.

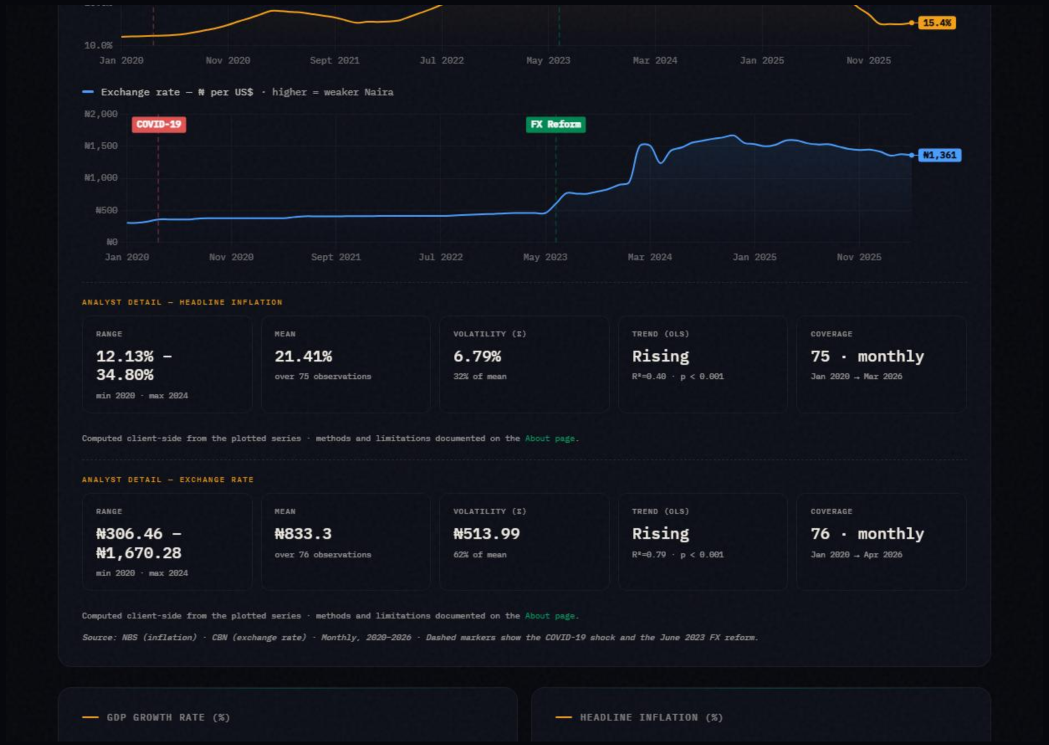


Figure 4.4 — The Reader/Analyst dial revealing statistical detail.

ANALYTICS

The analytics engine

Inferential, not just descriptive — strength and reliability reported together.

- **Full profile** — latest, YoY, range, volatility, trend.
- **Correlation + significance** — Pearson r with R^2 and a p-value.
- **Correlation matrix** — 12 indicators, cross-correlated at once.
- **Standardise & project** — z-scores + OLS trend.
- **No black box** — classical, checkable, reproducible.

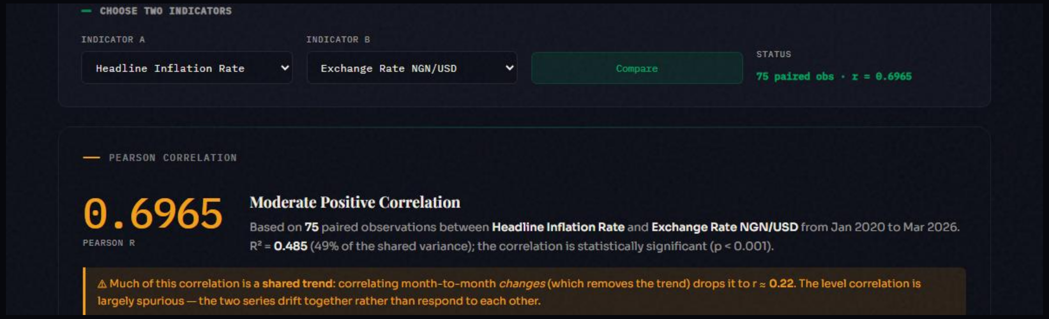


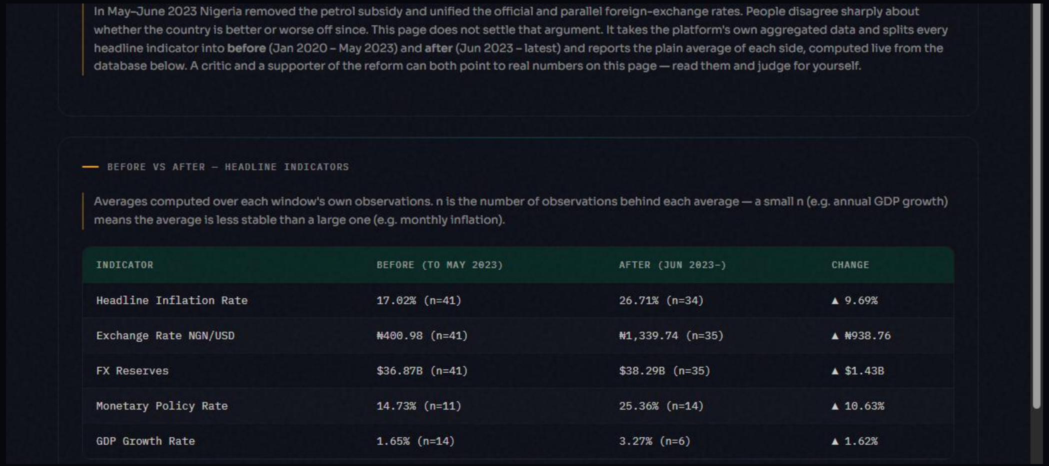
Figure 4.9 — Correlation reported with R^2 and a significance p-value.

IN ACTION

Reform Impact — takes no side

The aggregation applied to a live debate, without taking a political side.

- **June 2023** — subsidy removal + FX unification, split before/after.
- **Computed live** — every headline indicator, both sides.
- **Neutral** — critic and supporter cite the same real numbers.


In May–June 2023 Nigeria removed the petrol subsidy and unified the official and parallel foreign-exchange rates. People disagree sharply about whether the country is better or worse off since. This page does not settle that argument. It takes the platform's own aggregated data and splits every headline indicator into **before** (Jan 2020 – May 2023) and **after** (Jun 2023 – latest) and reports the plain average of each side, computed live from the database below. A critic and a supporter of the reform can both point to real numbers on this page — read them and judge for yourself.

BEFORE VS AFTER — HEADLINE INDICATORS

Averages computed over each window's own observations. n is the number of observations behind each average — a small n (e.g. annual GDP growth) means the average is less stable than a large one (e.g. monthly inflation).

INDICATOR	BEFORE (TO MAY 2023)	AFTER (JUN 2023–)	CHANGE
Headline Inflation Rate	17.02% (n=41)	26.71% (n=34)	▲ 9.69%
Exchange Rate NGN/USD	₦400.98 (n=41)	₦1,339.74 (n=35)	▲ ₦938.76
FX Reserves	$36.87B (n=41)	$38.29B (n=35)	▲ $1.43B
Monetary Policy Rate	14.73% (n=11)	25.36% (n=14)	▲ 10.63%
GDP Growth Rate	1.65% (n=14)	3.27% (n=6)	▲ 1.62%

Figure 4.15 — Reform Impact: before/after June 2023, neutral readings.

INTEGRITY

It warns you when it might mislead

The differentiator: a tool that flags when a result may be misleading.

- **Spurious-correlation guard** — level r vs detrended r.
- **Significance ≠ strength** — kept visually separate.
- **Validation as a service** — the Playground — and it never writes.

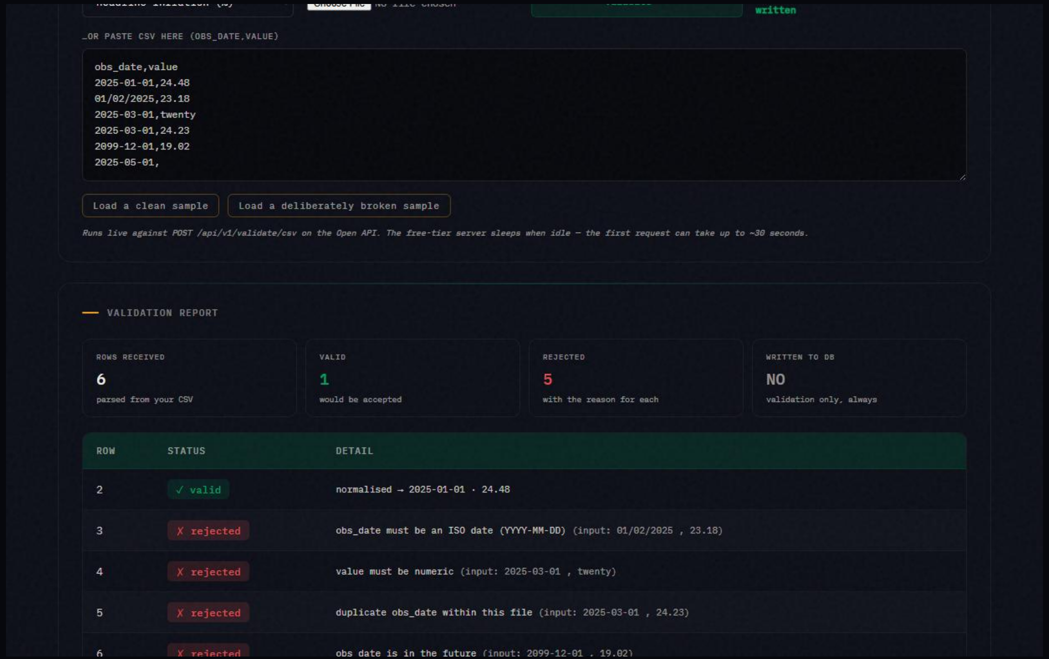


Figure 4.12 — Pipeline Playground: per-row verdicts, nothing written.

TESTING

Checked, not eyeballed

Correctness is demonstrated, not asserted.

- **24 automated tests** — endpoints, HATEOAS, cache — all pass.
- **Stats validated** — p-values vs known cases.
- **Truth audit** — found and fixed real defects.
- **Robust** — survives NaN / ∞ and 5,000-row floods.

RESULTS

What it delivers

Concrete, verifiable outputs mapped to the objectives.

- **122 indicators / 12,100 obs** — one queryable store.
- **17 live endpoints** — HATEOAS Level 3.
- **100 / 100** — accessibility, zero build.
- **Audited correct** — figures verified against data.

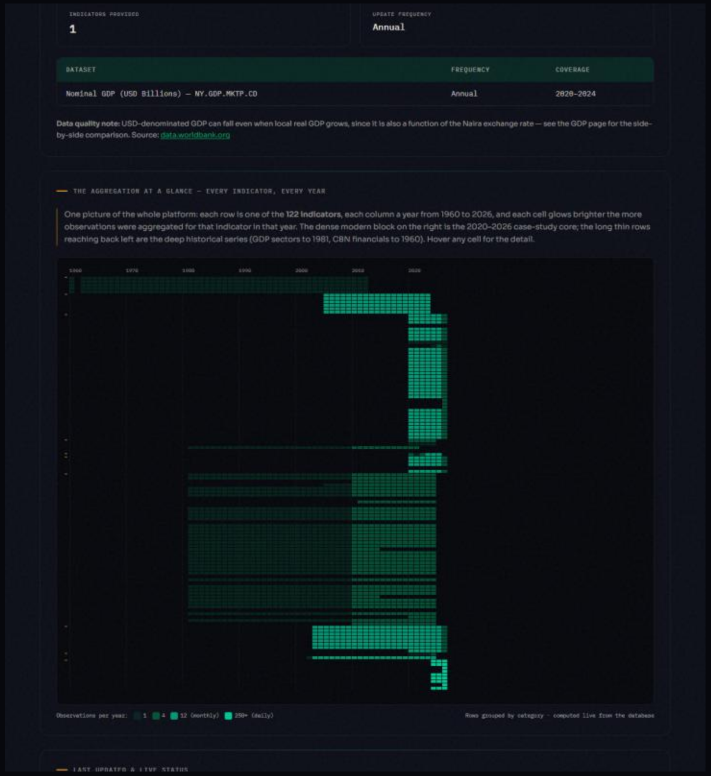


Figure 4.10 — The aggregation at a glance: 122 indicators × 1960–2026.

CONTRIBUTION

What's new

A reusable artefact and an argument, both reproducible from the repository.

- **A reference implementation** — of unified Nigerian economic data.
- **A genuinely open API** — no freely accessible equivalent found.
- **All free tools** — reproducible from the repo alone.
- **A governance argument** — it was always a choice, not a barrier.

WHAT'S NEXT

Where it goes

Known limitations, each paired with a concrete next step.

- **Automate collection** — scheduled connectors.
- **Expand coverage** — states, longer history.
- **Richer analytics** — seasonal adjustment, forecasting.
- **SDKs & auth tiers** — for heavier API users.

Conclusion

Nigeria's scattered, non-machine-readable public economic data need not stay that way: it can be consolidated into a correct, programmatically usable resource using only free and open-source tools.

Dashboard: antd-cr7.github.io/nigerian-dashboard
Open API: npedata-api.onrender.com (docs at /docs)
Source: github.com/ANTD-CR7/nigerian-dashboard

Thank you. Questions & live demonstration.